IIT HYDERABAD

Department of Electrical Engineering

Electronic Devices and Circuits Laboratory Report

Experiment No: **06**



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Contents

1 Objective **3**

2 Components and Equipment Required **3**

3 Circuit Configuration **3**

4 Theory **4**
4.1 Open-Loop Differential Amplifier 4
4.2 Derivation of the Feedback Factor 4
4.3 Unity-Gain Feedback Factor 5
4.4 Gain-of-10 Feedback Factor 5
4.5 Loop Gain 6
4.6 Derivation of the Closed-Loop Gain 6
4.7 Ideal Closed-Loop Gain 7

5 Theoretical Calculations **8**
5.1 Unity-Gain Configuration 8
5.2 Gain-of-10 Configuration 9

6 Theoretical Predictions **10**
6.1 Theoretical Cutoff Frequency 10

7 Low-Frequency Response **11**
7.1 Unity-Gain Configuration 11
7.2 Gain-of-10 Configuration 11

8 Observations **12**
8.1 Instrument Readings 13
8.2 Closed-Loop Gain at 1 kHz 14
8.2.1 Unity-gain configuration 15
8.2.2 Gain-of-10 configuration 15
8.3 Frequency Response 15
8.3.1 Unity-Gain Configuration 16
8.3.2 Gain-of-10 Configuration 16

9 Simulations **16**
9.1 Unity Gain Configuration 17
9.2 Gain of 10 Configuration 18

10 Results and Analysis **19**
10.1 Comparison of Measured and Predicted Gain 19
10.1.1 Unity-Gain Configuration 19
10.1.2 Gain-of-10 Configuration 20
10.2 Back-Calculation of Open-Loop Gain 21

11 Graphs **21**
11.1 Gain versus Frequency 21
11.2 Comparison of Closed-Loop Gains 22

12 Discussion **22**

13 Conclusion **23**

1 Objective

The objective of this experiment is to apply negative feedback around the Experiment 5 two-transistor MOSFET differential amplifier using external resistor-capacitor networks.

The differential amplifier is configured first as a unity-gain stage and then as a gain-of-10 stage. The closed-loop gain is measured and compared with the ideal feedback prediction based on the resistor ratio and with the finite-loop-gain prediction using the experimentally measured open-loop gain from Experiment 5.

The low-frequency roll-off produced by the AC-coupling capacitors is also investigated by measuring the frequency response of both configurations.

2 Components and Equipment Required

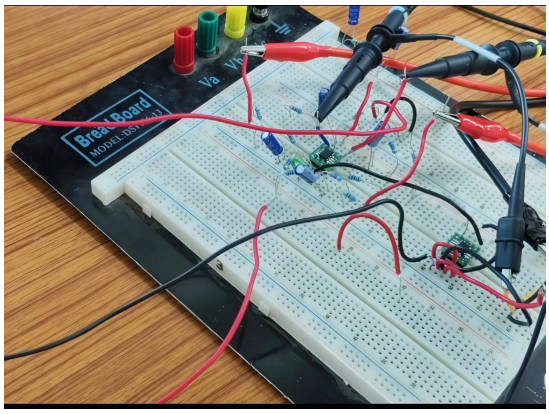
Table 1: Components and equipment used

Component	**Specification**	**Qty.**
Experiment 5 circuit	**SI9926BDY** dual N-channel MOSFET differential amplifier	1
Feedback resistor R_f	$10\,\text{k}\Omega$, $\pm 5\%$	2
Input resistor R_{in}	$10\,\text{k}\Omega$, $\pm 5\%$ (unity gain) ; $1\,\text{k}\Omega$, $\pm 5\%$ (gain of 10)	2 each
Feedback capacitor C_f	$1\,\mu\text{F}$	2
Input capacitor C_{in}	$1\,\mu\text{F}$ (unity gain) ; $10\,\mu\text{F}$ (gain of 10)	2 each
Bias resistor R_{bias}	$1\,\text{M}\Omega$	2
Function generator	Sine input	1
Oscilloscope	Two-channel oscilloscope	1

3 Circuit Configuration

The differential amplifier from Experiment 5 is used without changing its DC bias network. The feedback network is connected in a crossed configuration, where each output is fed back to the opposite input through an $R_f - C_f$ branch to provide negative feedback.

The input and feedback networks are AC-coupled using capacitors, while the $1\,\text{M}\Omega$ resistors maintain the original DC bias and isolate the feedback network from the bias potentiometers.



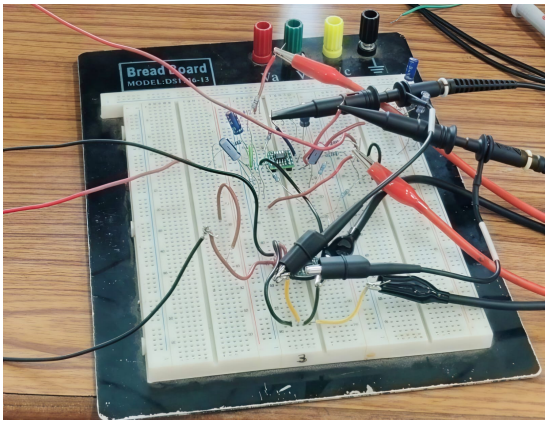


Figure 1: Differential Amplifier with crossed negative feedback

4 Theory

4.1 Open-Loop Differential Amplifier

Let A_d represent the open-loop differential voltage gain of the differential amplifier.

The open-loop relationship can be written as

$$V_o = -A_d V_i, \tag{1}$$

where V_i is the differential voltage applied to the amplifier input and V_o is the differential output voltage.

The negative sign arises from the crossed feedback configuration used in this experiment.

From Experiment 5,

$$\boxed{A_d = 49.8\,\mathrm{V/V}.} \tag{2}$$

4.2 Derivation of the Feedback Factor

The feedback network consists of R_f and R_{in}.

Since the MOSFET gate has a very high input impedance, the current entering the gate is approximately zero. Therefore, R_f and R_{in} form a simple voltage-divider network.

Let V_f be the voltage fed back to the input.

The voltage divider gives

$$V_f = V_o \frac{R_{in}}{R_f + R_{in}}. \tag{3}$$

The feedback factor β is defined as the fraction of the output voltage that is fed back to the input:

$$\beta = \frac{V_f}{V_o}. \tag{4}$$

Substituting the voltage-divider expression,

$$\boxed{\beta = \frac{R_{in}}{R_f + R_{in}}} \tag{5}$$

Thus, β depends only on the resistor values of the feedback network.

4.3 Unity-Gain Feedback Factor

For the unity-gain configuration,

$$R_f = 10\,\text{k}\Omega, \qquad R_{in} = 10\,\text{k}\Omega. \tag{6}$$

Therefore,

$$\beta = \frac{10\,\text{k}\Omega}{10\,\text{k}\Omega + 10\,\text{k}\Omega} \tag{7}$$

$$= \frac{10}{20} \tag{8}$$

$$= 0.5. \tag{9}$$

Hence,

$$\boxed{\beta = 0.5} \tag{10}$$

Therefore, 50% of the output is fed back through the feedback network.

4.4 Gain-of-10 Feedback Factor

For the gain-of-10 configuration,

$$R_f = 10\,\text{k}\Omega, \qquad R_{in} = 1\,\text{k}\Omega. \tag{11}$$

Therefore,

$$\beta = \frac{1\,\text{k}\Omega}{10\,\text{k}\Omega + 1\,\text{k}\Omega} \tag{12}$$

$$= \frac{1}{11} \tag{13}$$

$$\approx 0.0909. \tag{14}$$

Thus,

$$\boxed{\beta \approx 0.0909} \tag{15}$$

Approximately 9.09% of the output voltage is fed back.

4.5 Loop Gain

The loop gain is the product of the open-loop gain and the feedback factor :

$$\boxed{T = A_d \beta} \tag{16}$$

Using

$$A_d = 49.8, \tag{17}$$

the loop gain for the unity-gain configuration is

$$T = 49.8(0.5) \tag{18}$$

$$= 24.9. \tag{19}$$

Therefore,

$$\boxed{T_{\text{unity}} = 24.9} \tag{20}$$

For the gain-of-10 configuration,

$$T = 49.8(0.0909) \tag{21}$$

$$\approx 4.53. \tag{22}$$

Therefore,

$$\boxed{T_{\text{gain}-10} \approx 4.53} \tag{23}$$

The loop gain is much smaller for the gain-of-10 configuration. Hence, the effect of the finite open-loop gain is expected to be more significant in the gain-of-10 configuration.

4.6 Derivation of the Closed-Loop Gain

Let V_s be the differential source voltage and V_i be the differential voltage actually appearing at the amplifier input.

Since the MOSFET gate draws negligible current, applying KCL at the input node gives

$$\frac{V_s - V_i}{R_{in}} = \frac{V_i - V_o}{R_f}. \tag{24}$$

Multiplying by $R_{in}R_f$,

$$R_f(V_s - V_i) = R_{in}(V_i - V_o). \tag{25}$$

Expanding,

$$R_f V_s - R_f V_i = R_{in} V_i - R_{in} V_o. \tag{26}$$

Rearranging,

$$R_f V_s + R_{in} V_o = (R_f + R_{in}) V_i. \tag{27}$$

For the crossed negative-feedback configuration,

$$V_o = -A_d V_i. \tag{28}$$

Therefore,

$$V_i = -\frac{V_o}{A_d}. \tag{29}$$

Substituting this relation into the KCL equation gives

$$R_f V_s = -\frac{R_f + R_{in}(1 + A_d)}{A_d} V_o. \tag{30}$$

Therefore,

$$\boxed{\frac{V_o}{V_s} = -\frac{A_d R_f}{R_f + R_{in}(1 + A_d)}} \tag{31}$$

Hence, the finite-loop-gain closed-loop voltage gain is

$$\boxed{A_v = -\frac{A_d R_f}{R_f + R_{in}(1 + A_d)}} \tag{32}$$

4.7 Ideal Closed-Loop Gain

When the loop gain is sufficiently large,

$$A_d \beta \gg 1, \tag{33}$$

the effect of the finite open-loop gain becomes small.

The closed-loop gain then approaches

$$\boxed{A_v \approx -\frac{R_f}{R_{in}}} \tag{34}$$

The negative sign represents the inversion produced by the crossed feedback configuration.

5 Theoretical Calculations

5.1 Unity-Gain Configuration

For the unity-gain configuration,

$$R_f = 10\,\text{k}\Omega, \qquad R_{in} = 10\,\text{k}\Omega. \tag{35}$$

The ideal gain is

$$A_{v,\text{ideal}} = -\frac{R_f}{R_{in}} \tag{36}$$

$$= -\frac{10\,\text{k}\Omega}{10\,\text{k}\Omega} \tag{37}$$

$$= -1. \tag{38}$$

Thus,

$$\boxed{A_{v,\text{ideal}} = -1} \tag{39}$$

Using the experimentally measured open-loop gain (From Experiment 5)

$$A_d = 49.8, \tag{40}$$

the finite-loop-gain prediction is

$$A_v = -\frac{49.8(10\,\text{k}\Omega)}{10\,\text{k}\Omega + 10\,\text{k}\Omega(1 + 49.8)} \tag{41}$$

$$= -\frac{49.8}{1 + 50.8} \tag{42}$$

$$= -\frac{49.8}{51.8} \tag{43}$$

$$\approx -0.961. \tag{44}$$

Therefore,

$$\boxed{A_{v,\text{predicted}} \approx -0.961} \tag{45}$$

The predicted gain magnitude is therefore

$$\boxed{|A_v| \approx 0.961} \tag{46}$$

5.2 Gain-of-10 Configuration

For the gain-of-10 configuration,

$$R_f = 10\,\text{k}\Omega, \qquad R_{in} = 1\,\text{k}\Omega. \tag{47}$$

The ideal gain is

$$A_{v,\text{ideal}} = -\frac{10\,\text{k}\Omega}{1\,\text{k}\Omega} \tag{48}$$

$$= -10. \tag{49}$$

Thus,

$$\boxed{A_{v,\text{ideal}} = -10} \tag{50}$$

Using $A_d = 49.8$, the finite-loop-gain prediction is

$$A_v = -\frac{49.8(10\,\text{k}\Omega)}{10\,\text{k}\Omega + 1\,\text{k}\Omega(1 + 49.8)} \tag{51}$$

$$= -\frac{498}{10 + 50.8} \tag{52}$$

$$= -\frac{498}{60.8} \tag{53}$$

$$\approx -8.19. \tag{54}$$

Therefore,

$$\boxed{A_{v,\text{predicted}} \approx -8.19} \tag{55}$$

and

$$\boxed{|A_v| \approx 8.19} \tag{56}$$

6 Theoretical Predictions

The measured open-loop differential gain from Experiment 5 is

$$A_d = 49.8\,\mathrm{V/V}. \tag{57}$$

The feedback factor is

$$\beta = \frac{R_{in}}{R_f + R_{in}}. \tag{58}$$

The loop gain is

$$T = A_d\beta. \tag{59}$$

The finite-loop-gain closed-loop gain is

$$A_v = -\frac{A_d R_f}{R_f + R_{in}(1 + A_d)}. \tag{60}$$

The theoretical values are summarized below.

Table 2: Theoretical closed-loop parameters

Configuration	R_f	R_{in}	β	$T = A_d\beta$	$A_{v,\mathrm{pred}}$
Unity gain	$10\,\mathrm{k\Omega}$	$10\,\mathrm{k\Omega}$	0.500	24.9	-0.961
Gain of 10	$10\,\mathrm{k\Omega}$	$1\,\mathrm{k\Omega}$	0.0909	4.53	-8.19

6.1 Theoretical Cutoff Frequency

Each input/feedback coupling network forms a high-pass RC network with

$$f_c = \frac{1}{2\pi RC}. \tag{61}$$

For the unity-gain configuration,

$$R = 10\,\mathrm{k\Omega}, \qquad C = 1\,\mu\mathrm{F}, \tag{62}$$

giving

$$f_c = \frac{1}{2\pi(10\,000)(1 \times 10^{-6})} \approx 15.9\,\mathrm{Hz}. \tag{63}$$

For the gain-of-10 configuration,

$$R = 1\,\mathrm{k\Omega}, \qquad C = 10\,\mu\mathrm{F}, \tag{64}$$

giving

$$f_c = \frac{1}{2\pi(1000)(10 \times 10^{-6})} \approx 15.9\,\text{Hz}. \tag{65}$$

Thus, both configurations have the same theoretical cutoff frequency:

$$\boxed{f_c \approx 15.9\,\text{Hz}} \tag{66}$$

7 Low-Frequency Response

Each coupling capacitor forms a high-pass network with the corresponding resistance.

The cutoff frequency is

$$\boxed{f_c = \frac{1}{2\pi RC}} \tag{67}$$

7.1 Unity-Gain Configuration

For the unity-gain input network,

$$R = 10\,\text{k}\Omega, \qquad C = 1\,\mu\text{F}. \tag{68}$$

Therefore,

$$f_c = \frac{1}{2\pi(10 \times 10^3)(1 \times 10^{-6})} \tag{69}$$

$$\approx 15.9\,\text{Hz}. \tag{70}$$

Hence,

$$\boxed{f_c \approx 15.9\,\text{Hz}} \tag{71}$$

7.2 Gain-of-10 Configuration

For the gain-of-10 input network,

$$R = 1\,\text{k}\Omega, \qquad C = 10\,\mu\text{F}. \tag{72}$$

Therefore,

$$f_c = \frac{1}{2\pi(1 \times 10^3)(10 \times 10^{-6})} \tag{73}$$

$$\approx 15.9\,\text{Hz}. \tag{74}$$

Thus,

$$\boxed{f_c \approx 15.9\,\text{Hz}} \tag{75}$$

The feedback network has

$$R_f = 10\,\text{k}\Omega, \qquad C_f = 1\,\mu\text{F}, \tag{76}$$

which also gives

$$f_c \approx 15.9\,\text{Hz}. \tag{77}$$

Therefore, both configurations are expected to exhibit approximately the same low-frequency cutoff.

8 Observations

8.1 Instrument Readings

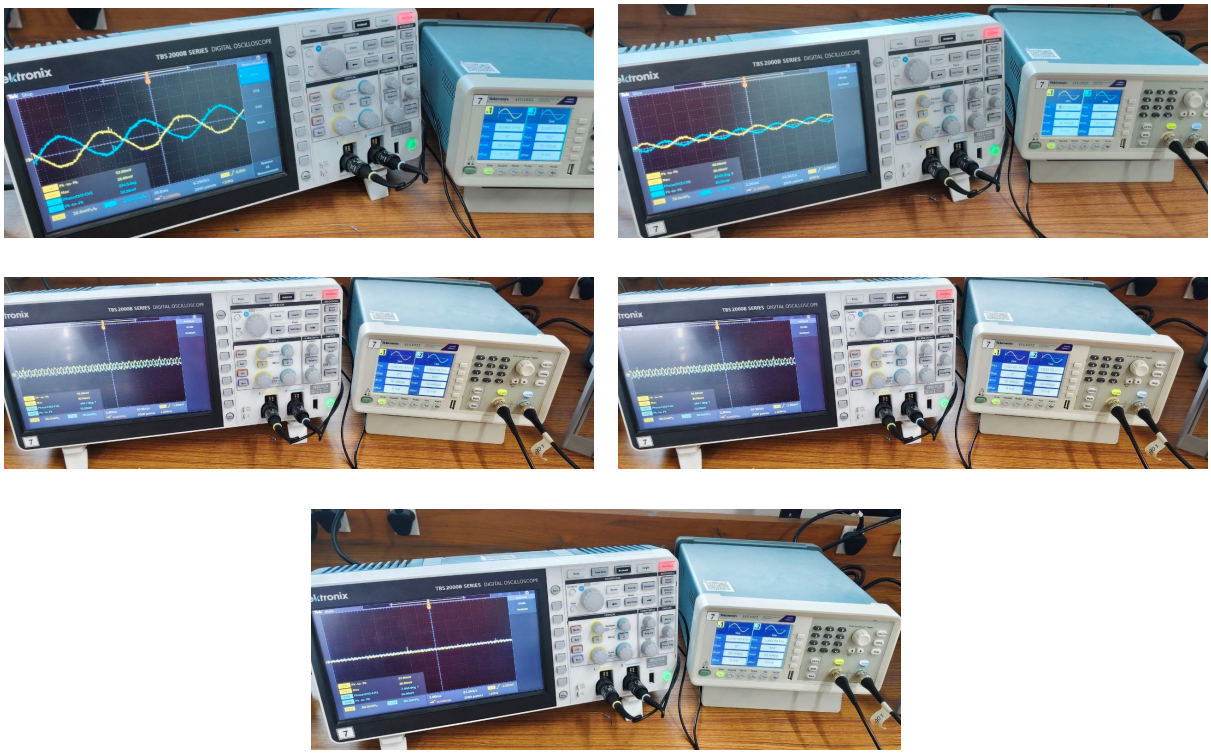


Figure 2: Unity Gain Configuration

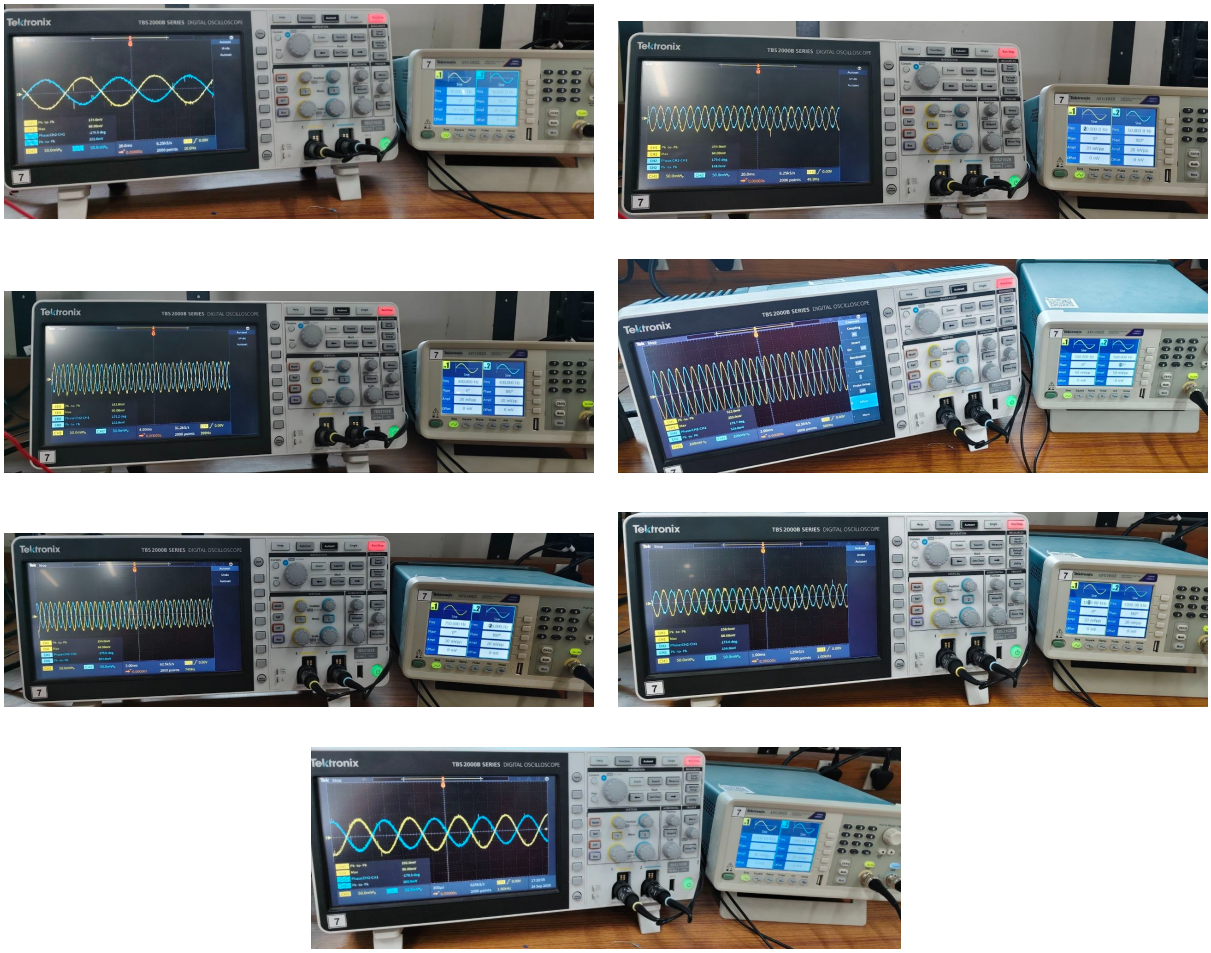


Figure 3: Gain of 10 Configuration

8.2 Closed-Loop Gain at 1 kHz

A frequency of 1 kHz was selected as the representative mid-band measurement frequency, as it lies within the approximately flat-gain region of the measured frequency response and is well above the theoretically estimated low-frequency cutoff.So,it is chosen as the mid-band frequency for evaluating the closed-loop gain of both configurations

The input voltage given in the experimental readings corresponds to the amplitude applied to each input branch. Hence, the differential input amplitude is taken as

$$V_{in,diff} = 2V_{in}. \tag{78}$$

Similarly, since the two drain outputs are approximately opposite in phase i.e, 180°, the differential output amplitude is obtained from

$$V_{out,diff} = V_{1out} + V_{2out}. \tag{79}$$

8.2.1 Unity-gain configuration

$$V_{in,diff} = 2(50\,\text{mV}) = 100\,\text{mV}. \tag{80}$$

At 1 kHz,

$$V_{out,diff} = 62\,\text{mV} + 52\,\text{mV} = 114\,\text{mV}. \tag{81}$$

Therefore,

$$|A_v| = \frac{114}{100} = 1.14. \tag{82}$$

8.2.2 Gain-of-10 configuration

$$V_{in,diff} = 2(20\,\text{mV}) = 40\,\text{mV}. \tag{83}$$

At 1 kHz,

$$V_{out,diff} = 158\,\text{mV} + 154\,\text{mV} = 312\,\text{mV}. \tag{84}$$

Thus,

$$|A_v| = \frac{312}{40} = 7.80. \tag{85}$$

The measured closed-loop gains are summarized below.

Table 3: Measured closed-loop gain at 1 kHz

Configuration	$V_{in,diff}$ **(mV)**	$V_{out,diff}$ **(mV)**	$\lvert A_v \rvert$
Unity gain	100	114	1.14
Gain of 10	40	312	7.80

8.3 Frequency Response

The measured output amplitudes for both configurations are given below. The gain magnitude is calculated from

$$\boxed{|A_v| = \frac{V_{out,diff}}{V_{in,diff}}} \tag{86}$$

and the corresponding gain in decibels is

$$\boxed{G_{\text{dB}} = 20\log_{10}|A_v|} \tag{87}$$

8.3.1 Unity-Gain Configuration

For this configuration,

$$V_{in,diff} = 100\,\text{mV}. \tag{88}$$

Table 4: Frequency response of the unity-gain configuration

f **(Hz)**	V_{1out} **(mV)**	V_{2out} **(mV)**	$V_{out,diff}$ **(mV)**	$\lvert A_v \rvert$	**Gain (dB)**
10	56.0	55.2	111.2	1.112	0.92
100	57.6	53.6	111.2	1.112	0.92
200	48.0	58.0	106.0	1.060	0.51
500	54.0	54.0	108.0	1.080	0.67
1000	62.0	52.0	114.0	1.140	1.14
2000	52.0	54.0	106.0	1.060	0.51

8.3.2 Gain-of-10 Configuration

For this configuration,

$$V_{in,diff} = 40\,\text{mV}. \tag{89}$$

Table 5: Frequency response of the gain-of-10 configuration

f **(Hz)**	V_{1out} **(mV)**	V_{2out} **(mV)**	$V_{out,diff}$ **(mV)**	$\lvert A_v \rvert$	**Gain (dB)**
10	174.0	152.0	326.0	8.15	18.22
50	154.0	148.0	302.0	7.55	17.56
400	182.0	152.0	334.0	8.35	18.43
500	204.0	204.0	408.0	10.20	20.17
750	194.0	164.0	358.0	8.95	19.04
1000	158.0	154.0	312.0	7.80	17.84
1500	192.0	180.0	372.0	9.30	19.37

9 Simulations

9.1 Unity Gain Configuration

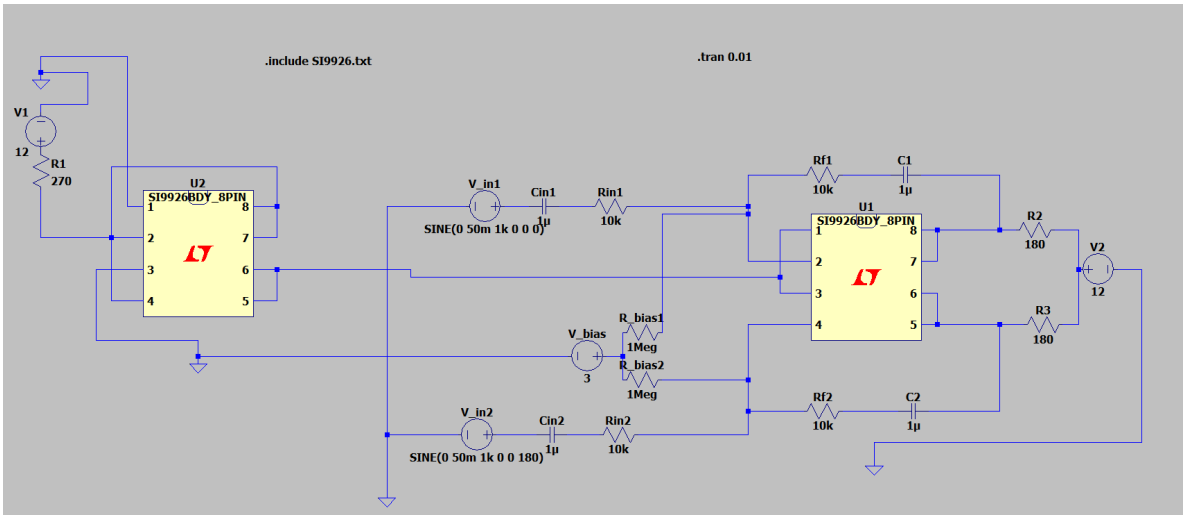


Figure 4: Unity Gain Configuration - Schematic

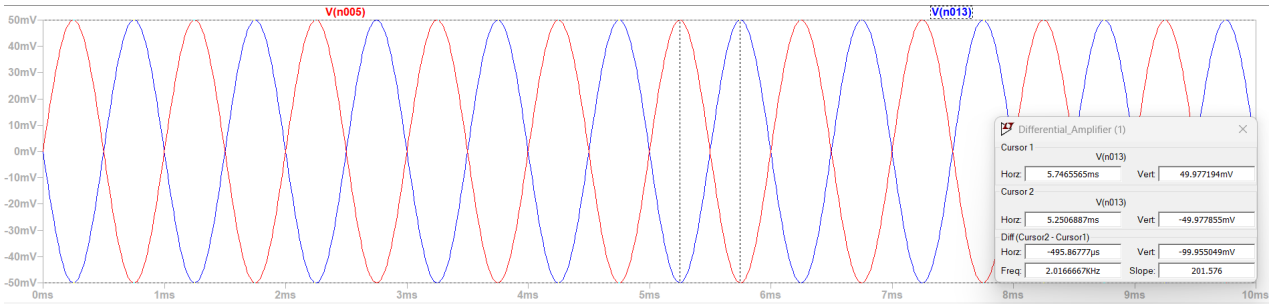


Figure 5: Small signal Input to the circuit

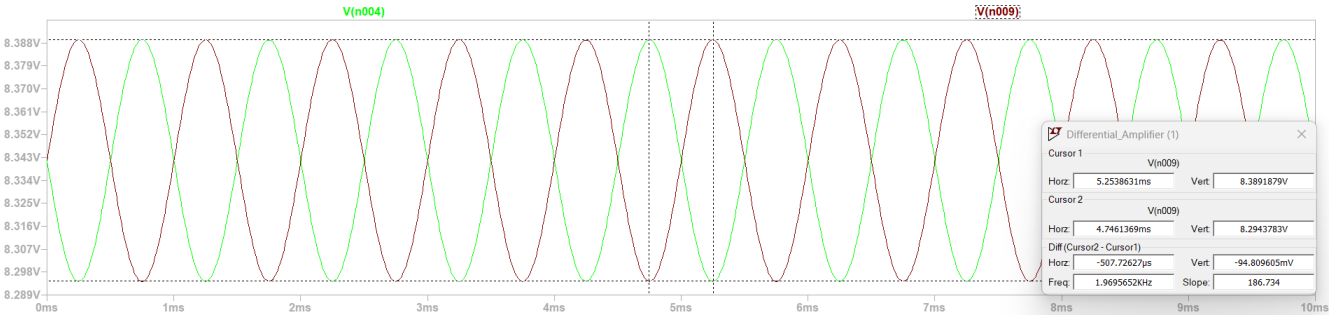


Figure 6: Output for the Input signal above

The input signal is a sinusoidal waveform with an amplitude of 50 mV and a frequency of 1 kHz. The output response is observed through simulation. From the waveform, the measured input and output amplitudes are approximately 99.95505 mV and 94.80961 mV, respectively. So the gain is calculated as

$$|A_v| = \left| \frac{94.80961 - (-94.80961)}{99.95505 - (-99.95505)} \right|$$

$$|A_v| = \frac{189.61922}{199.91010}$$

$$|A_v| \approx 0.9485$$

(Since the simulated dual MOSFETs are identical, the corresponding input and output voltages are equal)

9.2 Gain of 10 Configuration

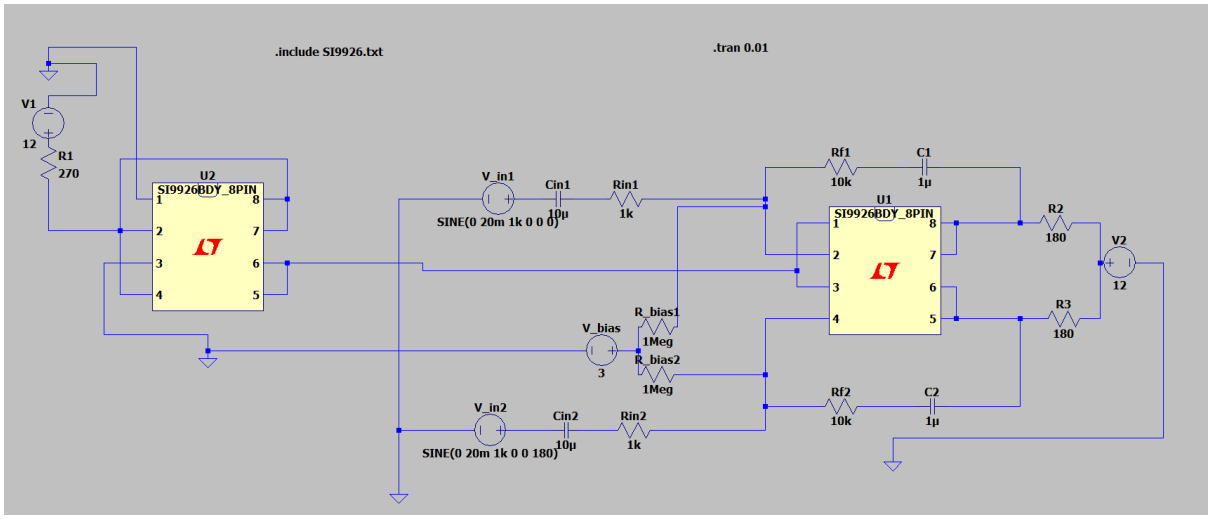


Figure 7: Gain of 10 Configuration - Schematic

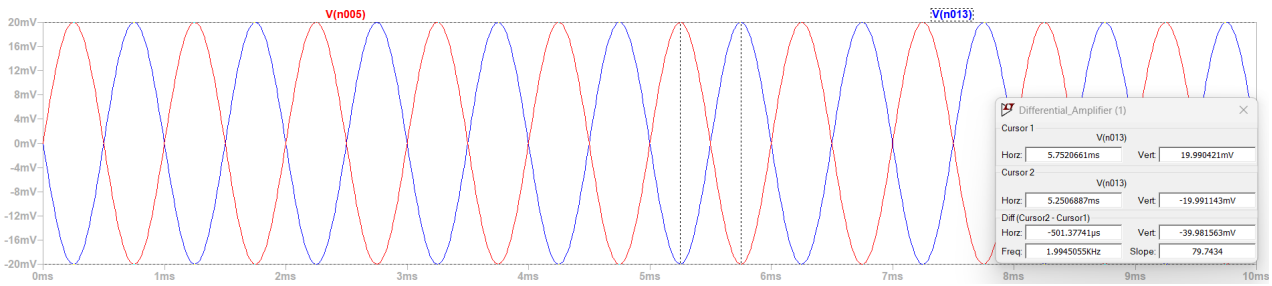


Figure 8: Small signal Input to the circuit

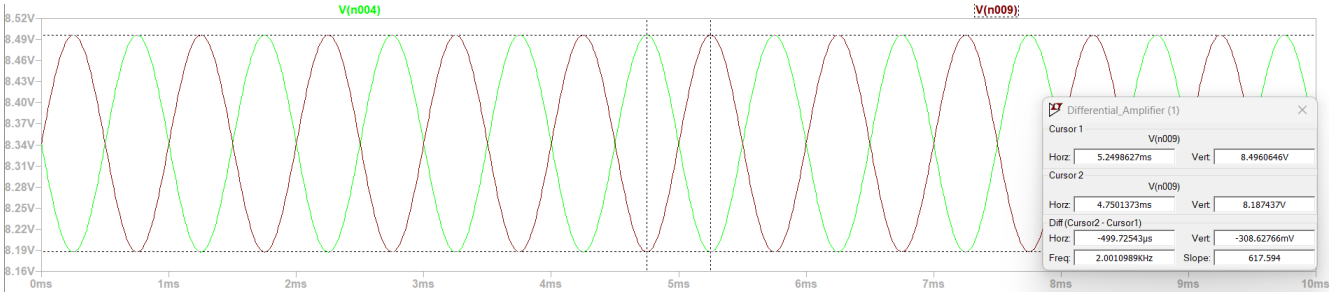


Figure 9: Output for the Input signal above

The input signal is a sinusoidal waveform with an amplitude of 20 mV and a frequency of 1 kHz. The output response is observed through simulation. From the waveform, the measured input and output amplitudes are approximately 39.981563 mV and 308.62766 mV, respectively. So the gain is calculated as

$$|A_v| = \left| \frac{308.62766 - (-308.62766)}{39.981563 - (-39.981563)} \right|$$

$$|A_v| = \frac{617.25532}{79.963126}$$

$$|A_v| \approx 7.7192$$

(Since the simulated dual MOSFETs are identical, the corresponding input and output voltages are equal)

10 Results and Analysis

10.1 Comparison of Measured and Predicted Gain

The percentage deviation from the finite-loop-gain prediction is calculated using

$$\boxed{\% \text{ deviation} = \frac{||A_{v,\text{measured}}| - |A_{v,\text{pred}}||}{|A_{v,\text{pred}}|} \times 100} \tag{90}$$

10.1.1 Unity-Gain Configuration

The measured gain at 1 kHz is

$$|A_{v,\text{measured}}| = 1.14. \tag{91}$$

Compared with the ideal gain,

$$\% \text{ deviation} = \frac{|1.14 - 1|}{1} \times 100 \tag{92}$$

$$= 14.0\%. \tag{93}$$

Compared with the finite-loop-gain prediction,

$$\% \text{ deviation} = \frac{|1.14 - 0.961|}{0.961} \times 100 \tag{94}$$

$$\approx 18.63\%. \tag{95}$$

10.1.2 Gain-of-10 Configuration

The measured gain at 1 kHz is

$$|A_{v,\text{measured}}| = 7.80. \tag{96}$$

Compared with the ideal gain,

$$\% \text{ deviation} = \frac{|7.80 - 10|}{10} \times 100 \tag{97}$$

$$= 22.0\%. \tag{98}$$

Compared with the finite-loop-gain prediction,

$$\% \text{ deviation} = \frac{|7.80 - 8.19|}{8.19} \times 100 \tag{99}$$

$$\approx 4.76\%. \tag{100}$$

The comparison is summarized below.

Table 6: Comparison of measured and theoretical gains

Configuration	**Ideal $\lvert A_v \rvert$**	**Predicted $\lvert A_v \rvert$**	**Measured $\lvert A_v \rvert$**	**Deviation from prediction**
Unity gain	1.00	0.961	1.14	18.63%
Gain of 10	10.00	8.19	7.80	4.76%

The gain-of-10 configuration shows better agreement with the finite-loop-gain model than with the ideal resistor-ratio prediction.

10.2 Back-Calculation of Open-Loop Gain

The open-loop gain can be estimated from the measured closed-loop gain using

$$A_d = \frac{|A_v|(R_f + R_{in})}{R_f - |A_v|R_{in}}. \tag{101}$$

For the gain-of-10 configuration,

$$|A_v| = 7.80, \qquad R_f = 10\,\text{k}\Omega, \qquad R_{in} = 1\,\text{k}\Omega. \tag{102}$$

Hence,

$$A_d = \frac{7.80(10+1)}{10 - 7.80} \tag{103}$$

$$= \frac{85.8}{2.2} \tag{104}$$

$$\approx 39.0\,\text{V/V}. \tag{105}$$

Thus, the back-calculated open-loop gain is

$$\boxed{A_d \approx 39.0\,\text{V/V}} \tag{106}$$

which is reasonably close to the Experiment 5 value of 49.8 V/V.

For the unity-gain configuration, the measured gain of 1.14 is greater than the ideal magnitude of 1. The finite-loop-gain model predicts a gain below unity for a positive finite open-loop gain. Therefore, substituting the measured unity-gain result into the back-calculation equation gives a non-physical value. This indicates that measurement uncertainty and non-ideal circuit behaviour are significant for this measurement.

Table 7: Back-calculated open-loop gain

Configuration	**Measured** $\lvert A_v \rvert$	**Back-calculated** A_d
Unity gain	1.14	Not physically meaningful
Gain of 10	7.80	39.0 V/V

11 Graphs

11.1 Gain versus Frequency

The measured gain in decibels is plotted against frequency using a logarithmic frequency axis.

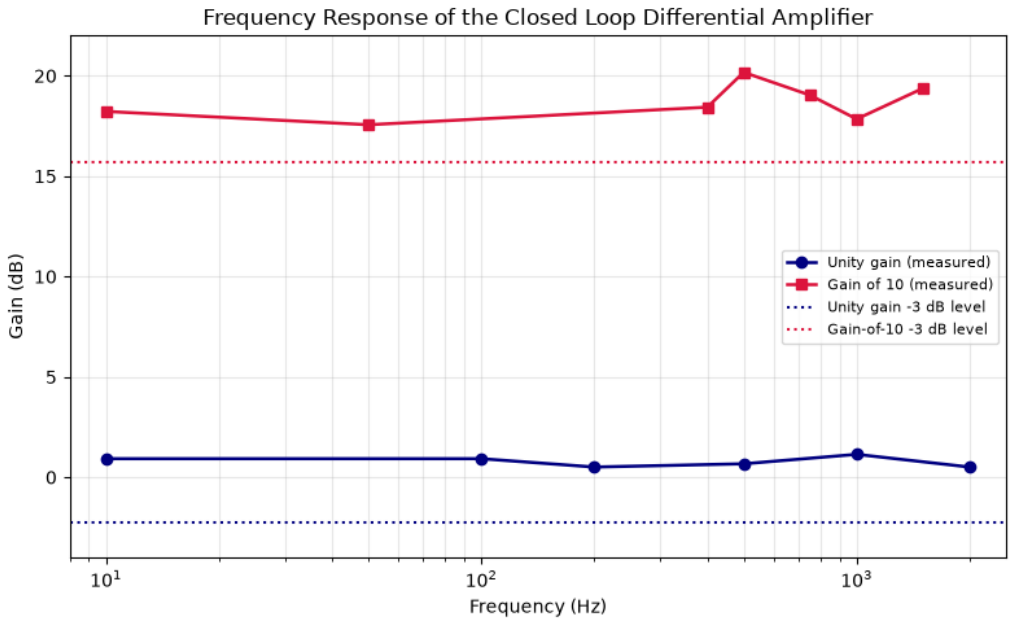


Figure 10: Gain(dB) vs Frequency

11.2 Comparison of Closed-Loop Gains

The measured gain is compared with the ideal and finite-loop-gain predictions.

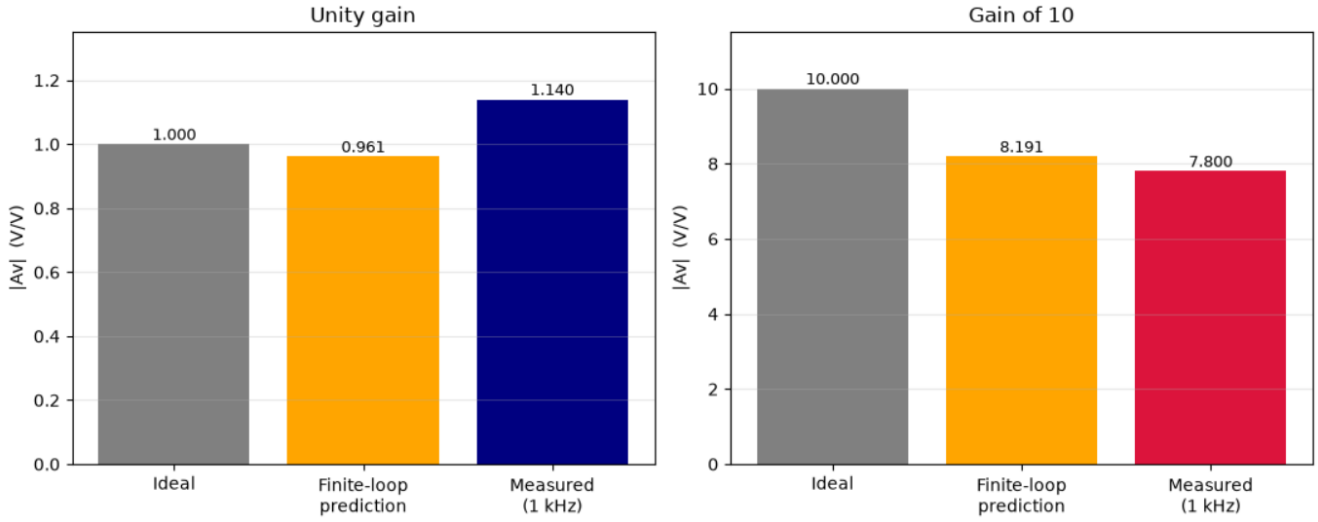


Figure 11: Comparison of measured and theoretical closed-loop gains

12 Discussion

The SI9926BDY differential amplifier was operated with crossed negative feedback in unity-gain and gain-of-10 configurations. The open-loop gain measured in Experiment 5, $A_d = 49.8\,\mathrm{V/V}$, was used for the predictions instead of assuming infinite gain. This gives loop gains of $T_{\mathrm{unity}} = 24.9$ and $T_{\mathrm{gain-10}} \approx 4.53$, so the finite open-loop gain matters more

in the gain-of-10 configuration.

For unity gain, the ideal gain is -1 and the predicted gain is -0.961. The measured gain of 1.14 is 18.63% above the prediction. Finite loop gain can only reduce the gain, so this excess must come from experimental error such as resistor tolerance ($\pm 5\%$) and oscilloscope reading error. The simulation gave 0.9485, within 1.3% of the prediction, which supports this.

For gain-of-10, the ideal gain is -10 and the predicted gain is -8.19. The measured gain of 7.80 is 4.76% below the prediction, and the simulated value of 7.72 agrees with it. Back-calculating the open-loop gain from this result gives $A_d \approx 39.0\,\mathrm{V/V}$, while the unity-gain result gives no physical value since $|A_v| > R_f/R_{in}$.

Feedback keeps the differential input of the pair at $V_{id} \approx V_{out,diff}/A_d$, about 2.3 mV for unity gain, far below the linear range of Experiment 5, which reduces distortion.Both configurations are expected to have a cutoff of $f_c = 1/(2\pi RC) \approx 15.9\,\mathrm{Hz}$. However, the measured gain stayed flat down to 10 Hz, so no -3 dB point was observed and the experimental cutoff could not be determined.

The differences between theory and experiment are mainly due to MOSFET mismatch, component tolerances, measurement uncertainty, and the simplicity of the feedback model.

13 Conclusion

The SI9926BDY differential amplifier was successfully operated with crossed negative feedback in unity-gain and gain-of-10 configurations. The measured gains were 1.14 and 7.80, compared with simulated values of 0.9485 and 7.72, respectively. The gain-of-10 result closely matched the theoretical prediction, while the unity-gain result showed slightly more deviation, mainly due to practical measurement and resistor tolerances.

Overall, the experiment demonstrated how negative feedback controls closed-loop gain and how finite open-loop gain and circuit non-idealities affect practical performance.